LERO Software for a better world

A World Leading SFI Research Centre

Science Foundation Ireland sfi For what's next

# CS5741
# Concurrency & Parallelism in Software Development

# Lecture 1: Key Terminology

Dr Faeq Alrimawi

HOST INSTITUTION

UNIVERSITY OF LIMERICK OLLSCOIL LUIMNIGH

PARTNER INSTITUTIONS

NUI Galway OÉ Gaillimh · Maynooth University · Trinity College Dublin, Coláiste na Tríonóide, Baile Átha Cliath, The University of Dublin · University College Dublin, An Coláiste Ollscoile, Baile Átha Cliath · UCC · MTU, Ollscoil Teicneolaíochta na Mumhan, Munster Technological University · DCU · Waterford Institute of Technology · Dundalk Institute of Technology, Dún Dealgan · LIT · GMIT

# Today, We Are Covering...

- Parallel Computing
- Concurrency
- Parallelism
- Concurrency vs. Parallelism
- Challenges
- Why Bother!?

# Parallel Computing

- It is **diving** a **problem** into **tasks** that can be executed at the same time.
  - The **goal** is to **reduce** the **time** needed to get a solution.
- **Problems** are getting more **complex**!
  - In Science: “grand challenge” problems (e.g., quantum physics, env. modelling).
  - AI, ML: large data and large models.
- Software & Hardware complement each other
  - Can’t have parallel computing without multiprocessors.
  - Also can’t have parallel computing without parallel programming that can utilise the multiprocessor hardware.

# SuperComputers

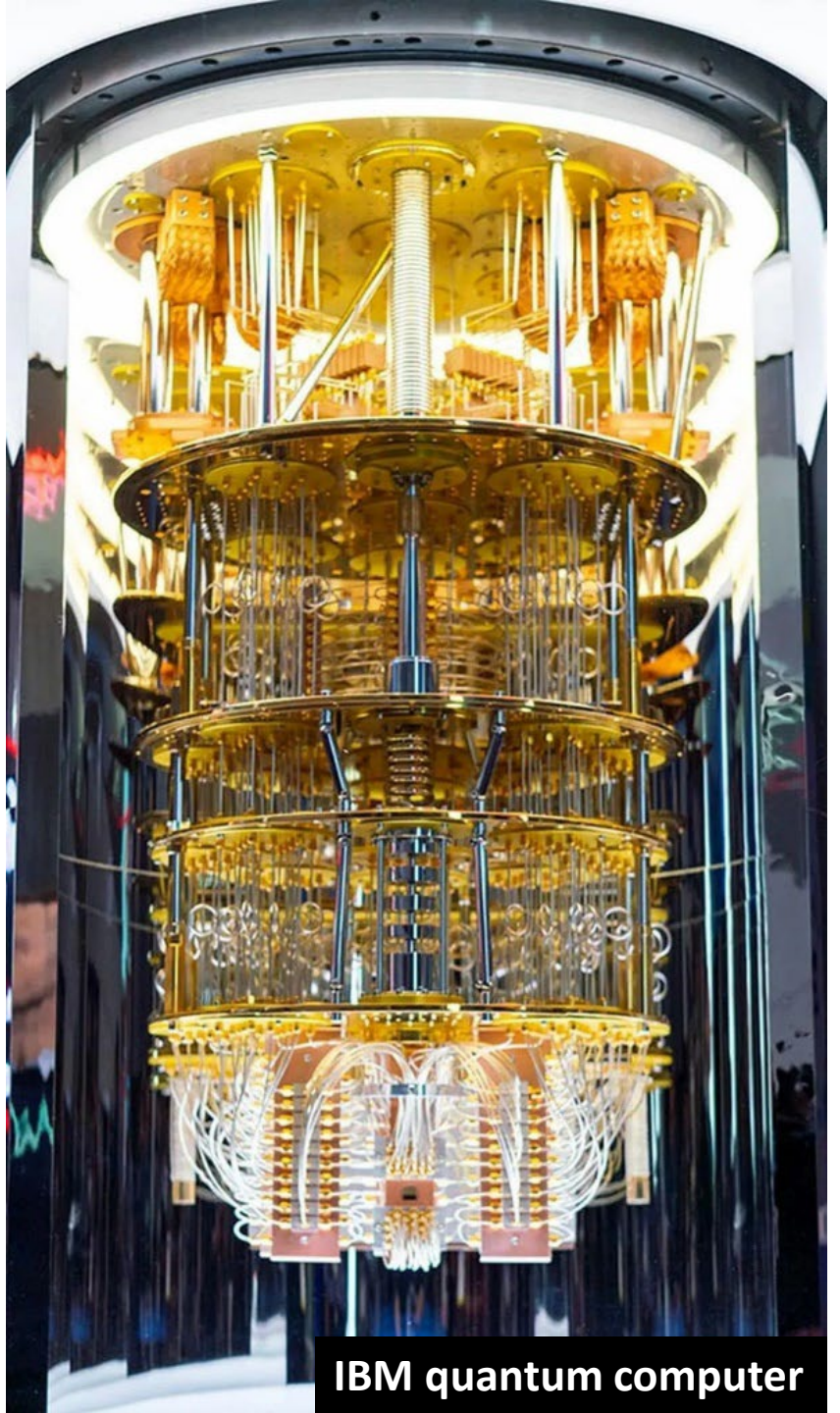



**~13.7million (CPU & GPU) cores**
**it can do ~2.7 quintillion calculations/s**

**LineShine, the most powerful supercomputer in the world right now!**

Src: https://top500.org/lists/top500/2026/06/

# Quantum Computers Next?



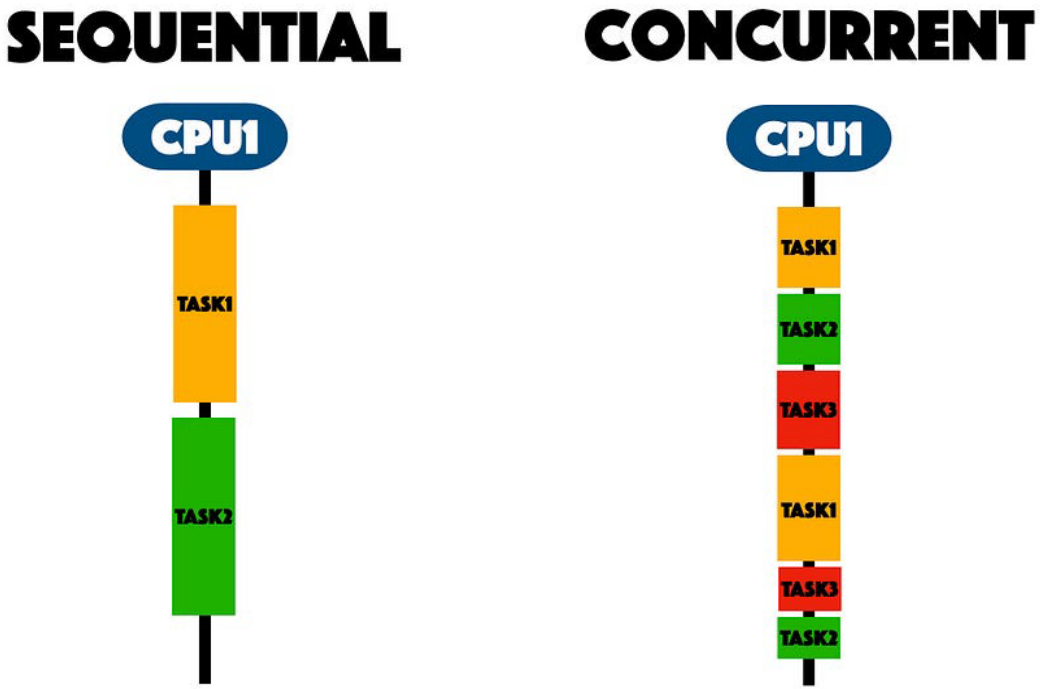
IBM quantum computer

# Today, We Are Covering...

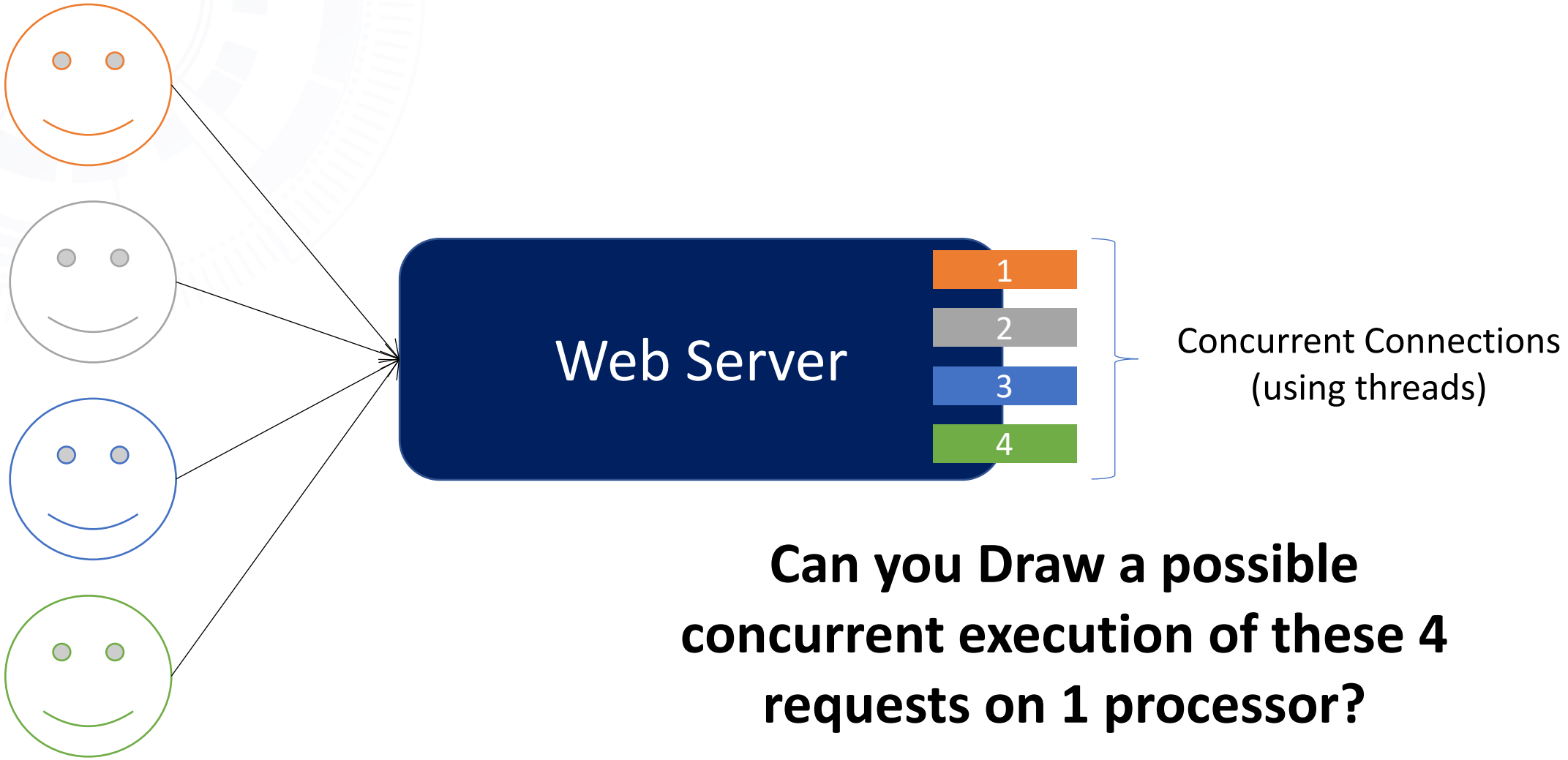
- Parallel Computing
- Concurrency
- Parallelism
- Concurrency vs. Parallelism
- Challenges
- Why Bother!?

# Concurrency

Progressing multiple tasks at the same time, where tasks may not necessarily run simultaneously but are interleaved

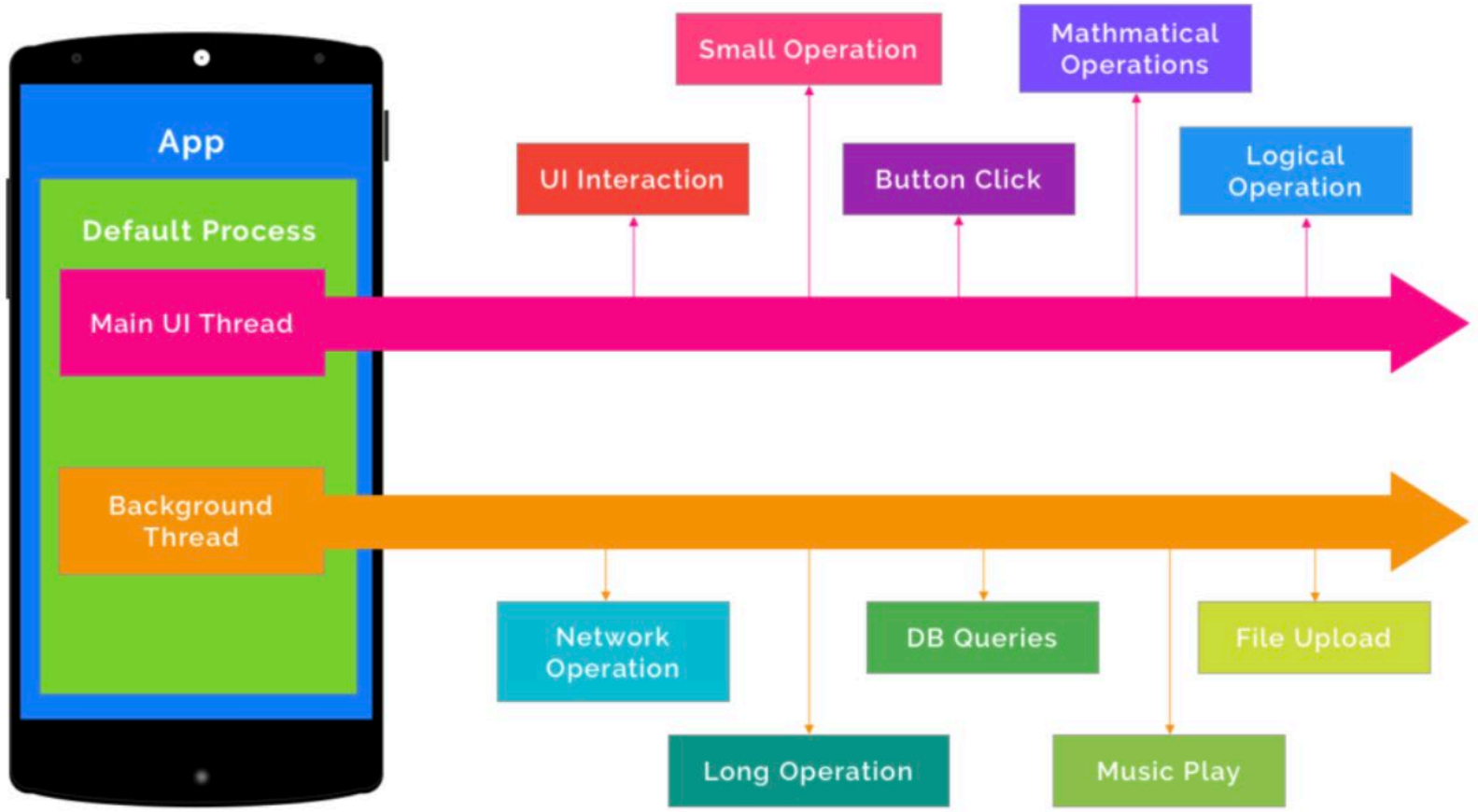


# Example 1: Web Server




**Can you Draw a possible concurrent execution of these 4 requests on 1 processor?**

# Example 2: UI Responsiveness



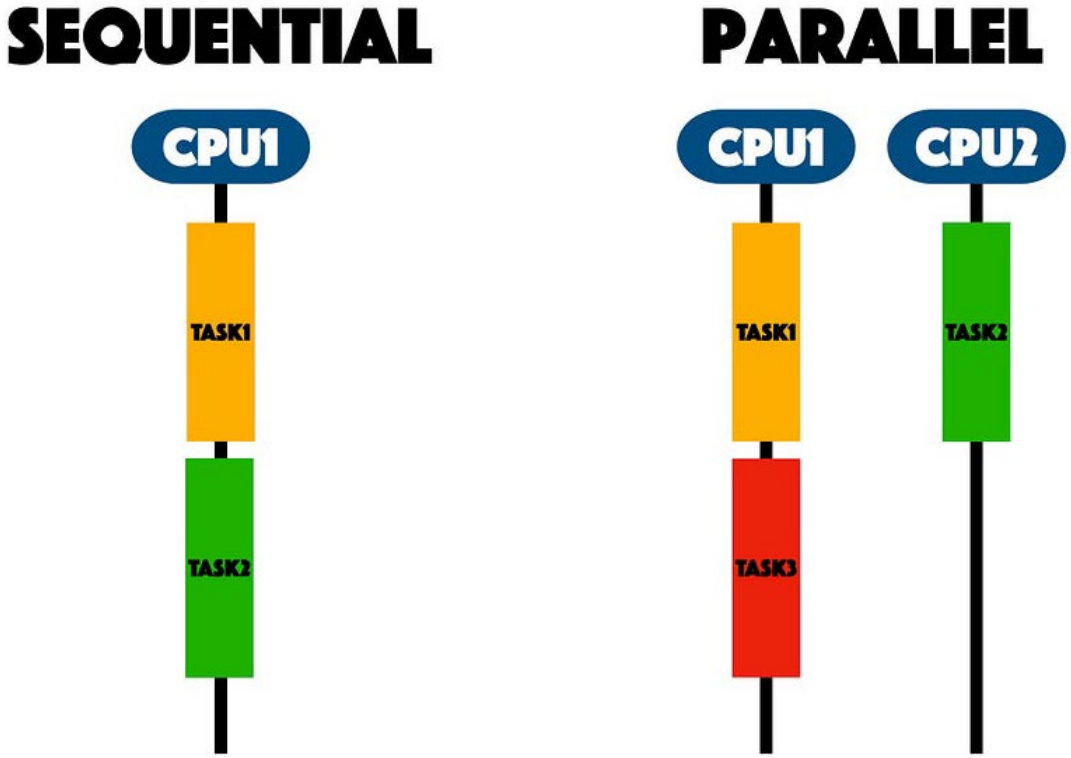

src: https://www.topcoder.com/thrive/articles/thread-in-android

# Today, We Are Covering…

- Parallel Computing
- Concurrency
- **Parallelism**
- Concurrency vs. Parallelism
- Challenges
- Why Bother!?

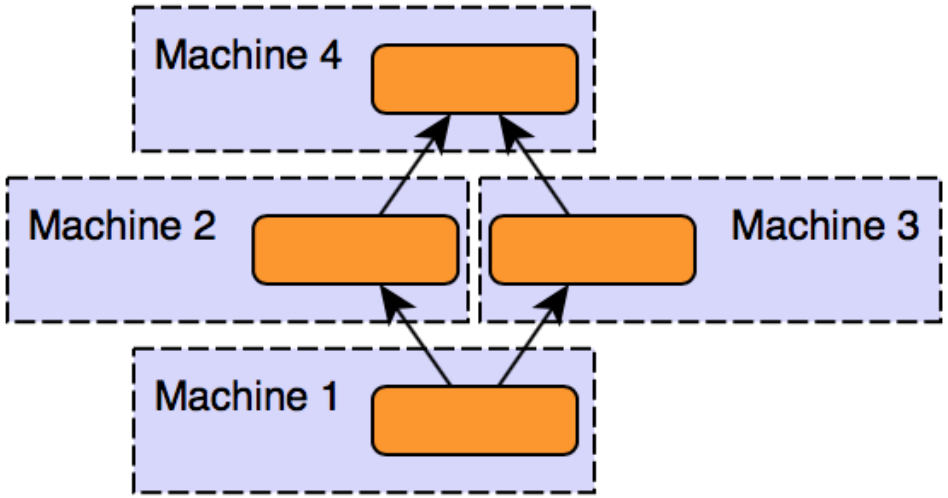
# Parallelism

Executing multiple tasks at the same time, often leveraging multi-core processors

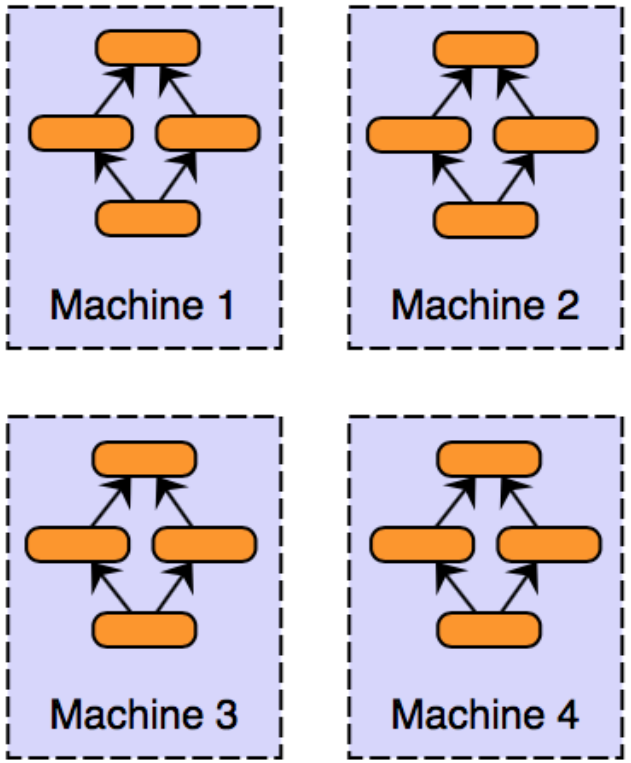


# Example 1: Machine Learning

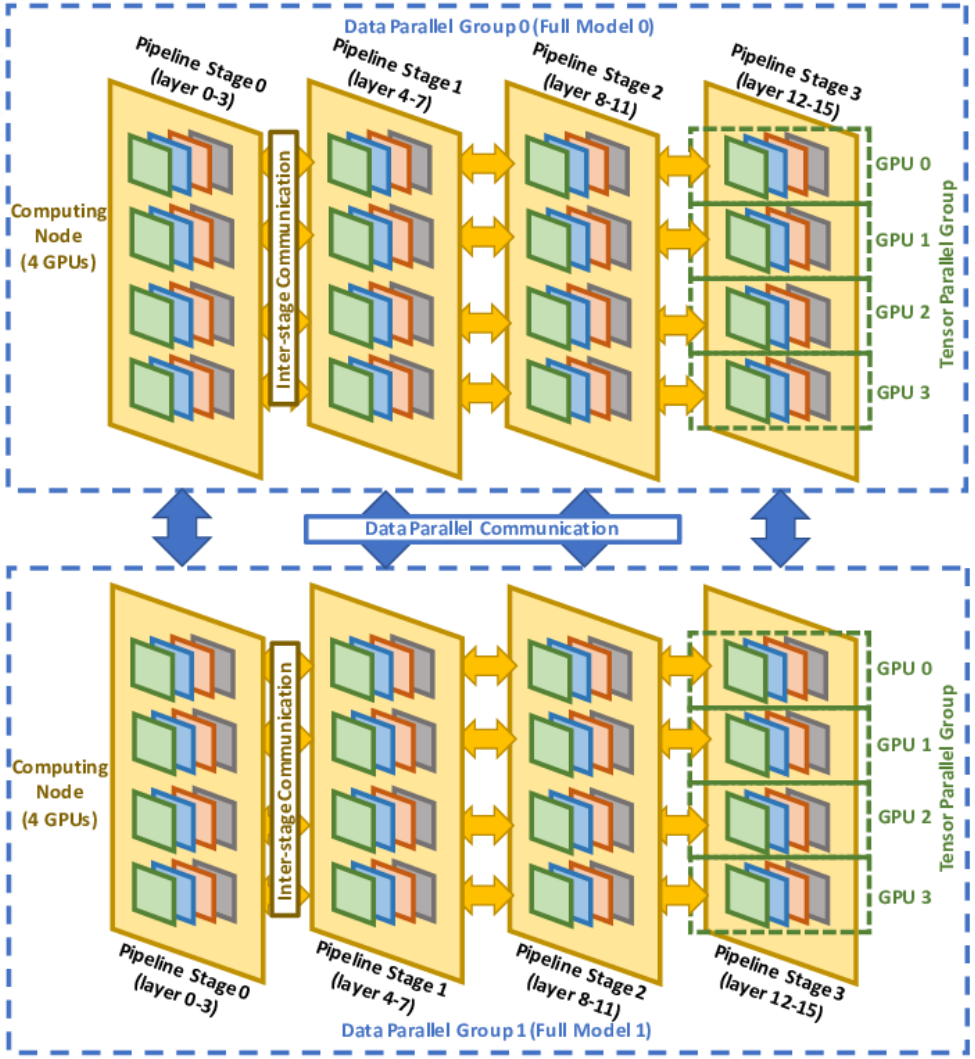
Model Parallelism




Data Parallelism



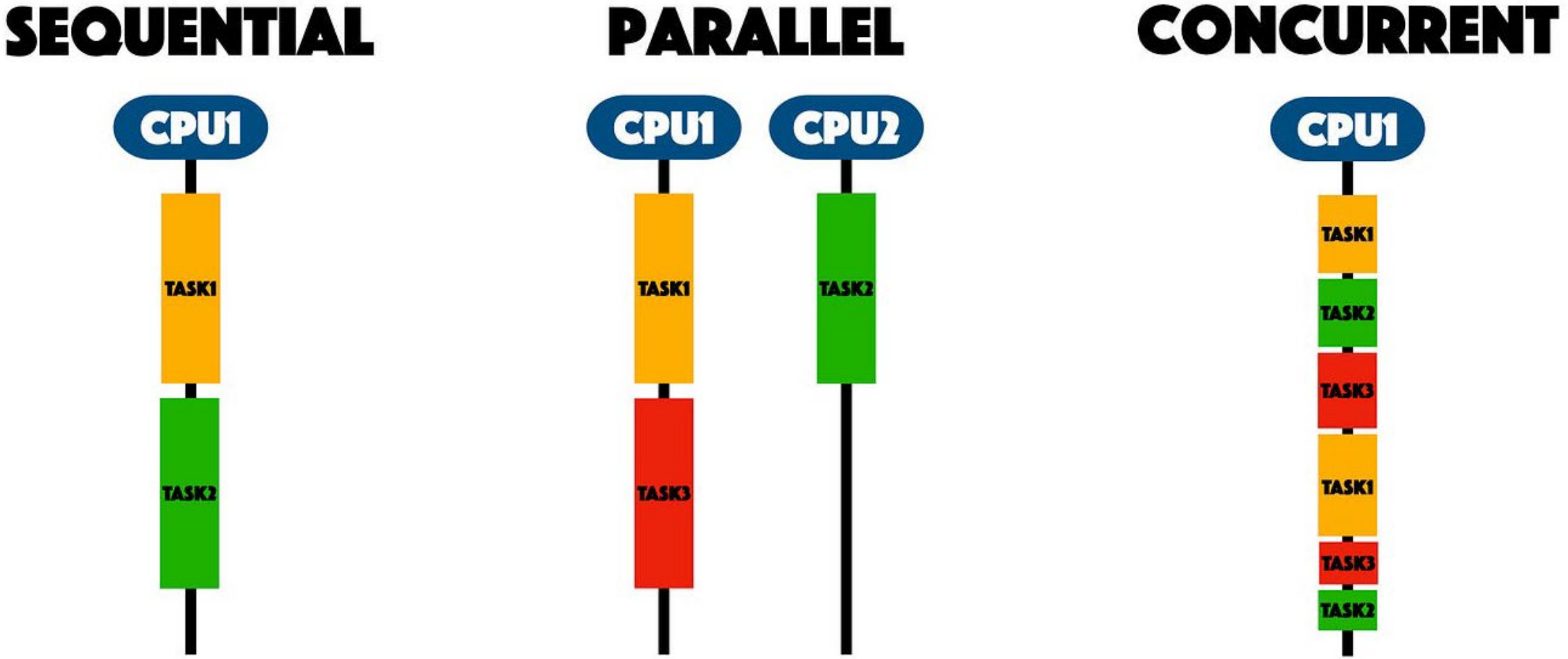
# Example 2: Graphics Rendering



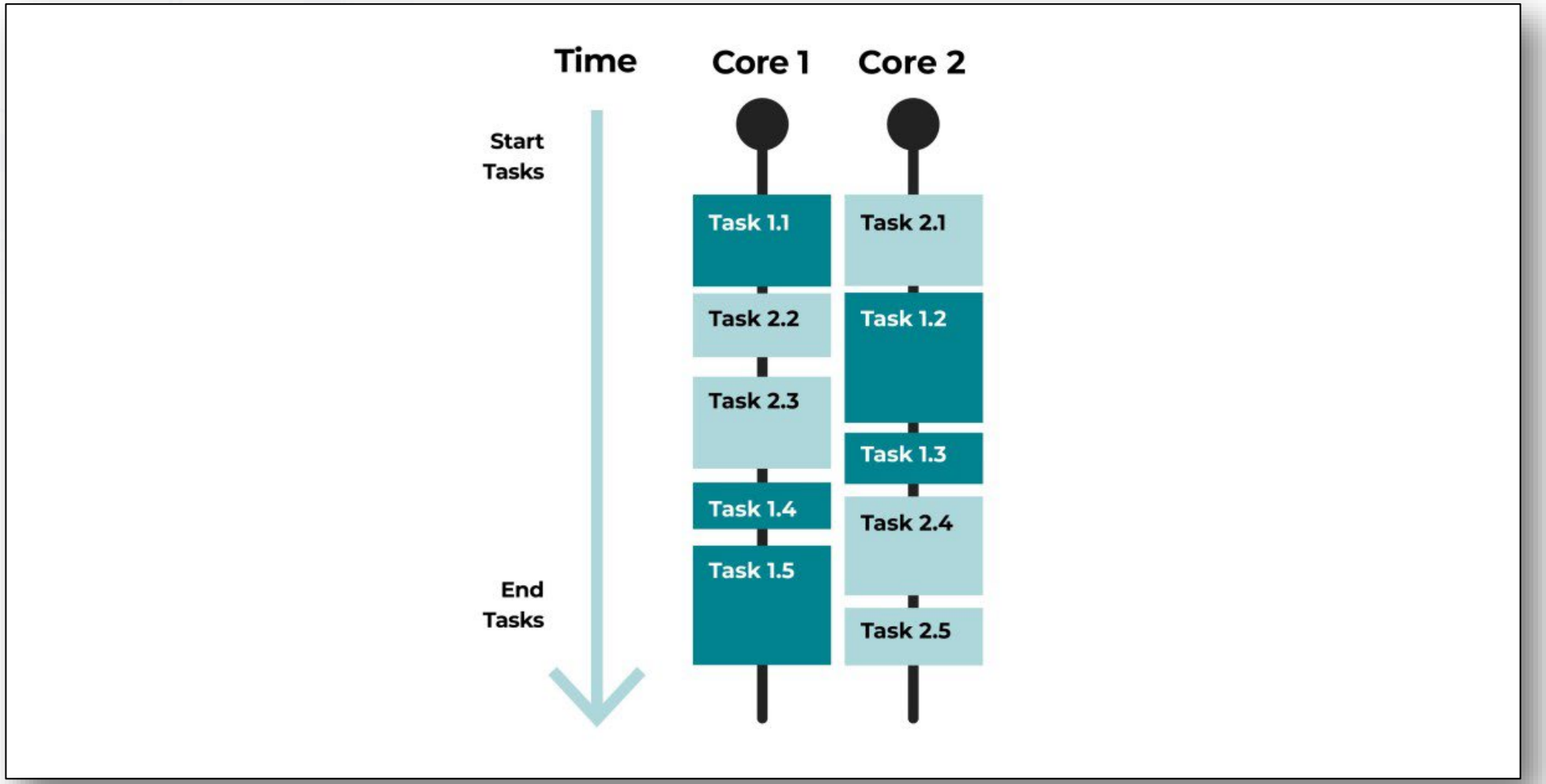
# Today, We Are Covering...

- Parallel Computing
- Concurrency
- Parallelism
- Concurrency vs. Parallelism
- Challenges
- Why Bother!?

# Concurrency vs. Parallelism (1)



# Concurrency Or Parallelism?



# Concurrency vs. Parallelism (2)

## Concurrency

- The ability to **progress** multiple tasks at the same time
- **Task Management:** Deals with **managing multiple tasks** within the same time frame
- **Execution:** Tasks may be **interleaved** but not executed simultaneously
- **Focus:** Making **efficient use of resources** by handling multiple tasks that can progress concurrently

## Parallelism

- The ability to **execute** multiple tasks at the same time
- **Simultaneous Execution**: Tasks are executed at the **same time** across different processors or cores.
- **Hardware Utilisation**: Requires **multi-core** or **multi-processor** systems to achieve true parallel execution.
- **Focus**: **Speeding up computations** and processing by running multiple tasks concurrently.

# Today, We Are Covering…

- Parallel Computing
- Concurrency
- Parallelism
- Concurrency vs. Parallelism
- Challenges
- Why Bother!?

# Challenges

## Partitioning

- Chopping your task into subtasks
- **Granularity**: finding the right size for subtasks
- **Load balancing**: ensuring the subtasks are evenly distributed over available resources

## Communication

- Exchanging data between subtasks
- **Latency**: the delay when data is communicated between subtasks
- **Overhead**: extra time and resources to transfer data (e.g., protocol overhead)

## Synchronisation

- Coordinating subtasks to ensure that they execute in a manner that maintains data **consistency** and **correctness**
- **Race conditions**: accessing shared resources without proper synchronisation
- **Deadlocks**: two or more tasks are waiting on each other to release resources

# Today, We Are Covering...

- Parallel Computing
- Concurrency
- Parallelism
- Concurrency vs. Parallelism
- Challenges
- Why Bother!?

# Why Bother!?

- Better **performance** and efficiency
- **Responsiveness** and User Experience
- Efficient Resource **Utilisation**

LERO Software for a better world

A World Leading SFI Research Centre

Science Foundation Ireland sfi For what's next

# What Are Your Questions?

HOST INSTITUTION

UNIVERSITY OF LIMERICK OLLSCOIL LUIMNIGH

PARTNER INSTITUTIONS

NUI Galway OÉ Gaillimh · Maynooth University · Trinity College Dublin, Coláiste na Tríonóide, Baile Átha Cliath, The University of Dublin · University College Dublin, An Coláiste Ollscoile, Baile Átha Cliath · UCC · MTU, Ollscoil Teicneolaíochta na Mumhan, Munster Technological University · DCU · Waterford Institute of Technology · Dundalk Institute of Technology, Institiúid Teicneolaíochta Dhún Dealgan · LIT · GMIT